

NETWORKS AND COMPLEXITY

Exercise Sheet 20: Motifs

*This is an exercise sheet from the forthcoming book *Networks and Complexity*.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 20.1: Multi-step adjacency [Lvl. 1]

Let λ_n be the eigenvalues of $\mathbf{A}$. Show that the eigenvalues of $\mathbf{A}^k$ are λ_n^k .

Ex 20.2: Zero power [Lvl. 1]

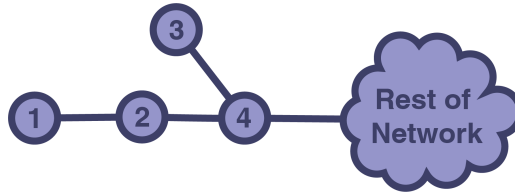
Consider $\sum \lambda_n^k$, where λ_n were the eigenvalues of the adjacency matrix $\mathbf{A}$. This sum reveals interesting network properties for several values of k . Is this also true for $k = 0$?

Ex 20.3: 5-chain [Lvl. 2]

Find the eigenvalues of the adjacency matrix of a chain of five nodes.

Ex 20.4: From the tip of the tail [Lvl. 2]

Consider the following network:



- Suppose the adjacency matrix has an eigenvector $\mathbf{v}$ with eigenvalue λ . Normalize the first element of this eigenvector to $v_1 = 1$, then determine the eigenvector elements v_2 , v_3 and v_4 . For which values of λ is $v_3 > 0$?
- Consider also the case where instead of the branch a chain of 10 nodes is attached to the rest of the network. Compute the eigenvector elements for the nodes in the chain for the case $\lambda = 9$.

Ex 20.5: Epidemic example revisited [Lvl. 3]

We revisit the epidemic model from Ex. 19.8. In this model the Jacobian of the disease-free state is $\mathbf{J} = p\mathbf{A} - r\mathbf{I}$, where p is the infection rate, r is the recovery rate $\mathbf{A}$ is the adjacency matrix and $\mathbf{I}$ is an identity matrix.

- Find the relationship between Jacobian eigenvalues κ_n and the adjacency eigenvalues λ_n .
- You notice that the network contains a fully-connected clique of six nodes. What does this tell you about the largest Jacobian eigenvalue κ_1 ?
- At which value of p will the clique destabilize the disease-free steady state? Is the transition that occurs at this point the epidemic threshold?

Ex 20.6: Competitive exclusion [Lvl. 3]

We consider an exploitative-competition (EC) motif in a food web: Our model includes two populations of predators, X_1 , X_2 , and a resource R , who are part of a large network that

is modeled with many additional variables. The dynamics of our three focal variables are described by

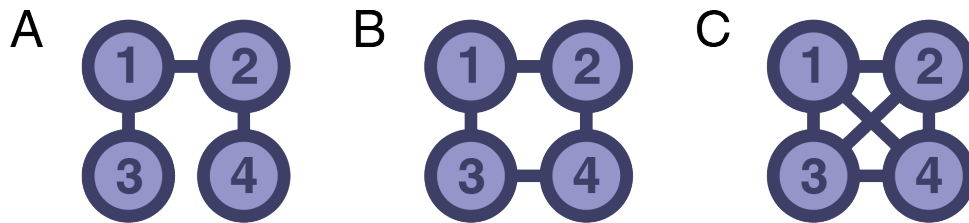
$$\begin{aligned}\dot{X}_1 &= g_1(R)X_1 - m_1X_1 \\ \dot{X}_2 &= g_2(R)X_2 - m_2X_2 \\ \dot{R} &= s(R) - g_1(R)X_1 - g_2(R)X_2 + \dots\end{aligned}$$

where g_1 , g_2 and s are arbitrary functions, m_1 and m_2 are arbitrary mortality rates for the two predator species, and ‘ $\dots$ ’ represents further terms that connect R to other variables.

- Formally compute the 3×3 -block of the Jacobian that describes the interaction between X_1 , X_2 , and R . Use the stationarity condition for X_1 and X_2 to simplify your matrix.
- Show that in any large ODE system that contains the EC motif studied here, one eigenvalue of the Jacobian will be zero.

Ex 20.7: Small eigenvalue puzzles [Lvl. 3]

Find the adjacency eigenvalues of the following networks:



Ex 20.8: The star and the snowflake [Lvl. 3]

Consider a node of degree 100, whose neighbors have degree 1.

- Find the eigenvalues and eigenvectors of the network’s adjacency matrix.
- Bonus: Take the 100 nodes of degree 1 and attach two nodes of degree 1 to each of them. Find all eigenvalues and eigenvectors.

Ex 20.9: Bipartite networks [Lvl. 4]

In this exercise we show that bipartite networks have a symmetric spectrum:

- Draw a bipartite simple graph with at least 5 nodes. Color the nodes with two colors such that there are no adjacent nodes with the same color.
- Now number the nodes starting with all nodes of the first color, followed by all nodes of the second color. Show that the adjacency matrix can be written as

$$\mathbf{A} = \begin{pmatrix} \mathbf{0} & \mathbf{B} \\ \mathbf{B}^\dagger & \mathbf{0} \end{pmatrix}$$

where $\mathbf{B}$ is a rectangular matrix.

- Assume we know an eigenvector $\mathbf{v}$ of $\mathbf{A}$ with eigenvalue λ . Write $\mathbf{v}$ as two shorter vectors $\mathbf{s}, \mathbf{r}$ stacked on top of each other:

$$\mathbf{v} = \begin{pmatrix} \mathbf{r} \\ \mathbf{s} \end{pmatrix}$$

Show that stacking $\mathbf{r}$ on $-\mathbf{s}$ yields an eigenvector with eigenvalue $-\lambda$.

- Now, prove that every bipartite network has a symmetric spectrum.

Ex 20.10: Reactivity [Lvl. 4]

Let $\mathbf{J}$ be the Jacobian of a steady state in an ODE system. Consider a small deviation $\boldsymbol{\delta}(t)$ from the steady state. Hence the absolute distance of the system from the steady state at time t is $|\boldsymbol{\delta}| = \sqrt{\boldsymbol{\delta}^T \boldsymbol{\delta}}$.

- a) Define a as the relative rate of change in $|\boldsymbol{\delta}|$, i.e. the current change in the distance normalized by the current distance, and show

$$a = \frac{1}{|\boldsymbol{\delta}|} \frac{d}{dt} |\boldsymbol{\delta}| = \frac{\boldsymbol{\delta}^\dagger \mathbf{H} \boldsymbol{\delta}}{\boldsymbol{\delta}^\dagger \boldsymbol{\delta}},$$

where $\mathbf{H} = (\mathbf{J} + \mathbf{J}^\dagger)/2$ is the Hermitian part of the Jacobian.

- b) Define the reactivity of the steady state as the highest possible value of a . Show that the reactivity is identical to the rightmost eigenvalue of $\mathbf{H}$.
c) Consider the predator prey system

$$\begin{aligned}\dot{X} &= X - XY/(3 + X) \\ \dot{Y} &= 2XY/(3 + X) - Y.\end{aligned}$$

Compute $\mathbf{J}$ and $\mathbf{H}$ and the reactivity in the nontrivial steady state.

- d) We now consider a larger system where the Jacobian $\mathbf{J}$ from (c) appears as a principal submatrix. (Perhaps the system contains X and Y along with many other variables). What can you say about the reactivity of the entire system?

Ex 20.11: More proofs [Lvl. 4]

Prove the following using direct computation, Gershgorin's theorem or Rayleigh quotient maximization, or eigenvalue properties:

- a) Regular graphs have an eigenvalue that is identical to their mean degree.
b) The highest node degree is an upper bound for the largest eigenvalue in any network.
c) The mean degree is a lower bound for the rightmost adjacency eigenvalue in any undirected simple graph.
d) Bipartite networks cannot contain cycles of odd length (prove this one algebraically, don't use the coloring intuition).

Ex 20.12: A large eigenvalue puzzle [Lvl. 5]

Find as many eigenvalues as possible of the dragon network:

